



APPLICATION NOTE

Evaluating Bosch Sensortec MEMS sensors over I3C with the Binho Supernova

AN0004

Document information

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Abstract	A MIPI I3C target implements a subset of the specification, and neither the datasheet nor the result code returned by a transfer reliably says which subset. This application note describes how to establish that subset by measurement, using a Binho Supernova USB host adapter and the Binho I3C Target Board to drive three Bosch Sensortec parts: the BMI323 inertial measurement unit and the BMP581 and BMP585 pressure sensors. It covers enumeration, identification, register access, in-band interrupts and bus rate limits, and supplies a reference Python utility that produces a support matrix from observable effects rather than from status codes. Measured results are given for all three parts.

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1

Introduction

A device datasheet lists the MIPI I3C common command codes the part supports, usually as a table with a column of Y and N entries. That table is the natural starting point when bringing up a new target, and on its own it is not sufficient to write a host against.

Two properties of I3C make it insufficient. The first is that a target which has been assigned a dynamic address acknowledges that address, and a target may acknowledge a command it does not implement and then discard it. Nothing distinguishes that outcome, at the host, from a command that was carried out. The second is that a CCC support table describes the command, not the effect: a part may recognize a command, answer it with a defined value, and still not do the thing the command names.

Both properties are visible in the three parts described here. All three answer at least ten commands their own datasheets mark as unsupported. One lists a command to enter a high-data-rate mode as supported and then explains in a footnote that high-data-rate operation is not supported at all. Two answer a query about their high-data-rate capability with a defined value meaning none.

None of this is a defect in the parts or in the documentation. It is a consequence of what I3C permits, and it means that a target's usable feature set has to be established by observing effects rather than by reading status. This document describes how to do that from a host, and supplies a utility that does it.

1.1

Scope

This document describes:

- enumeration and identification of an I3C target, and how to read a capability directly from the target's bus characteristics register
- the register access framing used by three Bosch Sensortec parts, which differs between them
- how to establish, by measurement, which common command codes a target implements
- configuring a sensor to raise its data-ready interrupt as an I3C in-band interrupt, with no interrupt pin connected
- the highest bus rates at which the parts operate without error, and the method used to establish them
- a reference host utility, supplied with this document, that performs all of the above

Sensor calibration, the algorithms in the parts' feature engines, and the vendor driver software for these devices are outside the scope of this document. The utility supplied here exercises the I3C target in a sensor; it is not a sensor driver.

1.2 Applicable devices

Table 1 lists the parts exercised for this document, and the parts the I3C Target Board accepts that were not exercised. A profile written from a datasheet alone is exactly the assumption this document argues against, so the untested rows are marked as such and no support claims are made for them.

Table 1. Bosch Sensortec parts and the basis for each row

Part	Function	Board	Basis
BMI323	6-axis IMU	Shuttle 3.0	Verified on hardware for this document
BMP581	Barometric pressure	Shuttle 3.0	Verified on hardware for this document
BMP585	Barometric pressure, media resistant	Shuttle 3.0	Verified on hardware for this document
BMM350	Magnetometer	Shuttle 3.0	Accepted by the I3C Target Board. Not exercised
BMA530, BMA580	Accelerometer	Shuttle 3.0	Accepted by the I3C Target Board. Not exercised

NOTE the method in Section 5.5 does not depend on the part. It applies to any I3C target, including the rows above that were not exercised and parts from other vendors.

1.3 Related documents

Table 2. Related documents

Document	Title
BST-BMI323-DS000	BMI323 datasheet
BST-BMP581-DS004	BMP581 datasheet
BST-BMP585-DS003	BMP585 datasheet
MIPI I3C v1.0	Improved Inter Integrated Circuit, 23 December 2016
MIPI I3C v1.1	Improved Inter Integrated Circuit, version 1.1
BIN103	Binho I3C Target Board product documentation

2 System overview

The host is a Binho Supernova USB host adapter acting as the I3C controller. The sensor is carried on a Bosch Shuttle 3.0 evaluation board, which plugs into the Binho I3C Target Board. The target board carries the bus between the Supernova's I3C connector and the shuttle

socket, and breaks out every target pin on 2.54 mm headers, with dedicated headers on SCL and SDA for a logic analyzer.

The signal path is two wires and a supply. No interrupt pin, reset line or strapping resistor is connected for any procedure in this document.

The I3C Target Board provides one Shuttle 3.0 socket and one STMicroelectronics DIL24 socket. The three parts covered here all use the Shuttle 3.0 socket, so they are covered one at a time, and the procedures below assume a single target on the bus. The bus is daisy-chainable and a reader may extend it; that configuration is not exercised in this document.

One property of the bus shapes everything that follows, and is stated here because the rest of the document is organized around it.

IMPORTANT a target that has been assigned a dynamic address acknowledges that address. A target may acknowledge a common command code it does not implement, discard it, and return nothing to distinguish that from having carried it out. A successful transfer is therefore evidence that the target was addressed, and is not evidence that the target supports the command. Section 5.5 describes the consequence.

3 Hardware setup

3.1 Required equipment

Table 3. Required equipment

Item	Note
Binho Supernova	Firmware 4.4.0 or later
Binho I3C Target Board	BIN103
Bosch Shuttle 3.0 board	One of BMI323, BMP581 or BMP585
USB-C cable	Host to Supernova

3.2 Connections

Table 4. Connections

From	To	Required
Supernova I3C connector	I3C Target Board I3C input	Yes
I3C Target Board Shuttle 3.0 socket	Bosch shuttle board	Yes
I3C Target Board power jumper	Supernova-supplied rail	Yes, unless an external supply is used
I3C Target Board SCL, SDA headers	Logic analyzer	No

The shuttle board is keyed to its socket. No sensor pin is wired individually for any procedure in this document.

3.3 Bus considerations

All measurements in this document were made at 3.3 V. The Supernova supplies the bus rail through the target board when the power jumper selects it.

CAUTION confirm the shuttle board's supply rating before applying a rail. A Shuttle 3.0 board carries its own regulators, and the sensor's interface supply is not necessarily the rail applied to the board.

The controller's two clock rates are configured separately and are not independent. The open-drain rate governs the broadcast phase of a transfer and the push-pull rate governs the data phase, and the adapter rejects some combinations as an invalid frequency pair. Section 8.2 gives the rates the parts sustained and the constraint observed between the two.

The starting values used throughout this document are 3.3 V, a push-pull rate of 5 MHz at 50% duty cycle, and an open-drain rate of 1 MHz. These are the utility's defaults.

4 Software setup

The utility requires Python 3.10 or later and the vendor SDK:

```
pip install binhosupernova
```

The versions used for the measurements in this document are given in Section 8.1. Extract the assets archive and run the utility from the extracted directory:

```
python i3c_mems.py scan
```

A single command confirms the whole path from host to sensor:

```

1 target(s) on the bus at 3300 mV, PUSH_PULL_5_MHZ_50_DC, OPEN_DRAIN_1_MHZ

dynamic address 0x08
  PID  07 70 10 51 10 00  device id 0x1051
      enumeration cached 07 70 10 51 00 00, which is the pre-settling value
  BCR  0x06  DCR 0x62
  HDR  SDR only  IBI capable
  profile bmp585

```

5 I3C on these devices

5.1 Enumeration

Bus initialization issues RSTDAA, which clears any dynamic address already assigned, followed by ENTDAAs, in which each target on the bus arbitrates using its 48-bit provisional identifier and is assigned a dynamic address. With one target present the assigned address is `0x08`.

ENTDAAs also captures each target's provisional identifier, bus characteristics register and device characteristics register into the controller's target table, which is why `scan` can report them without issuing any further command.

Assigning addresses at run time also relaxes a constraint that applies to these parts on a static-address bus. The BMP581 datasheet gives its I2C address as `0x46` for SDO at logic 0 and `0x47` for SDO at logic 1, and the BMP585 uses the same pair. Two such parts therefore share a static-address bus without difficulty, one at each address, but a third cannot be addressed at all. Under ENTDAAs the number of parts sharing an address pair on paper stops being a constraint, because each is assigned a distinct dynamic address during enumeration.

NOTE the address limit above is taken from the BMP581 and BMP585 datasheets and was not exercised for this document. The I3C Target Board has one Shuttle 3.0 socket, so the two parts were never on the bus at the same time. ENTDAAs itself was exercised on every part individually, which is what Section 6.1 documents.

NOTE the identifier captured during ENTDAAs is captured before the target has settled. On the BMP581 and BMP585, bit 12 of the provisional identifier reflects the level on the SDO pin, and it reads 0 in the cached copy and 1 when read back over the bus. Measured over eight consecutive enumerations, GETPID returned bit 12 as 0 on the first read every time and as 1 from the third read onward every time. The device identifier field is unaffected. The datasheets prescribe the remedy directly: a dummy read should be the first access to the device. The supplied utility discards two reads after enumeration for this reason, and `scan` reports both values where they differ.

5.2 Identity

Three registers identify a target, and all three are read with a single CCC each. Table 5 gives the values measured for the three parts.

Table 5. Identity values measured for the three parts

Part	PID	BCR	DCR	Chip ID register	Chip ID
BMI323	07 70 10 43 00 00	0x06	0xEF	0x00	0x43
BMP581	07 70 10 50 10 00	0x06	0x62	0x01	0x50
BMP585	07 70 10 51 10 00	0x06	0x62	0x01	0x51

Every value matches the corresponding datasheet. The device identifier field, bits 31 to 16 of the provisional identifier, embeds the chip identifier in each case: 0x1043, 0x1050 and 0x1051.

The BMP581 and BMP585 both report a DCR of 0x62. The MIPI device characteristics register registry assigns that value to a pressure sensor, so a controller can classify a target it has no prior knowledge of.

IMPORTANT the BMI323 chip identifier register is 16 bits wide and its upper byte is a reserved field that reads 0x11 on the parts measured, so the register reads 0x1143 while the datasheet gives a reset value of 0x0043. A host comparing the whole word against the datasheet value concludes the part is wrong. Compare the low byte.

5.3 Reading a capability from the target

Bit 5 of the bus characteristics register is the MIPI advanced capabilities bit. The BMI323 datasheet states its meaning for that part without ambiguity: bit 5 is set to 0b0, that is, the device supports solely SDR.

All three parts report a BCR of 0x06, in which bit 5 is clear. That single bit is the reliable answer to whether a target offers a high-data-rate mode, and it is available from any target on any bus in one command.

The alternatives are both unreliable, and Section 5.5 explains why:

- the CCC support table in the BMI323 datasheet marks ENTHDR0 as supported, and a footnote to the same table states that ENTHDR0 is recognized but the device does not support HDR operations
- the BMP581 and BMP585 support the CCC that reports HDR capability, and answer it with 0x00, meaning no HDR modes
- an attempted HDR-DDR transfer returns a successful result on all three parts, and the bus remains usable afterwards, so the attempt distinguishes nothing

Table 6 decodes the BCR value common to all three parts.

Table 6. BCR 0x06 decoded

Bits	Field	Value
7:6	Device role	I3C target
5	Advanced capabilities	Not supported, so SDR only
4	Virtual target support	Not supported
3	Offline capable	Always responds to bus commands
2	IBI mandatory payload	An accepted IBI is followed by a mandatory byte
1	IBI capable	Yes
0	Max data speed limit	No limit declared

5.4 Register access

The three parts do not share a register access model, and the differences are not a vendor convention that can be assumed once and reused. Table 7 gives what was measured.

Table 7. Register access measured per part

Part	Register address	Data width	Dummy bytes before a read payload
BMI323	8 bits	16 bits	2
BMP581	8 bits	8 bits	0
BMP585	8 bits	8 bits	0

The BMI323 datasheet states the dummy byte count for each interface, giving two for I3C. The BMP581 and BMP585 datasheets do not state a count for I3C, and none was observed.

Where the count is not documented, it can be established by reading a run of registers whose reset values are known and finding the offset at which they line up. Two adjacent anchors are enough to remove ambiguity. On the BMP585, CHIP_ID at `0x01` reads `0x51` and REV_ID at `0x02` reads `0x32`:

```
dummy=0: raw 51 32      -> CHIP_ID 0x51 REV_ID 0x32  <== both anchors match
dummy=1: raw 51 32 22    -> CHIP_ID 0x32 REV_ID 0x22
dummy=2: raw 51 32 22 00 -> CHIP_ID 0x22 REV_ID 0x00
```

A single anchor would have admitted more than one offset.

The broadcast address may be omitted from a transfer. Reads issued with and without the leading `0x7E` header returned the same value on all three parts, which matches the BMP581 and BMP585 datasheets: the host may skip the 7E header and start with the dynamic address section, and this applies to all I3C transactions.

5.5 What "supported" means on this bus

The BMI323 datasheet marks ten common command codes as not supported: SETMWL, SETMRL, GETMWL, GETMRL, GETMXDS, GETHDRCAP, ENTAS1, ENTAS2, ENTAS3 and SETGRPA. Every one of them returns a successful result when sent to the part.

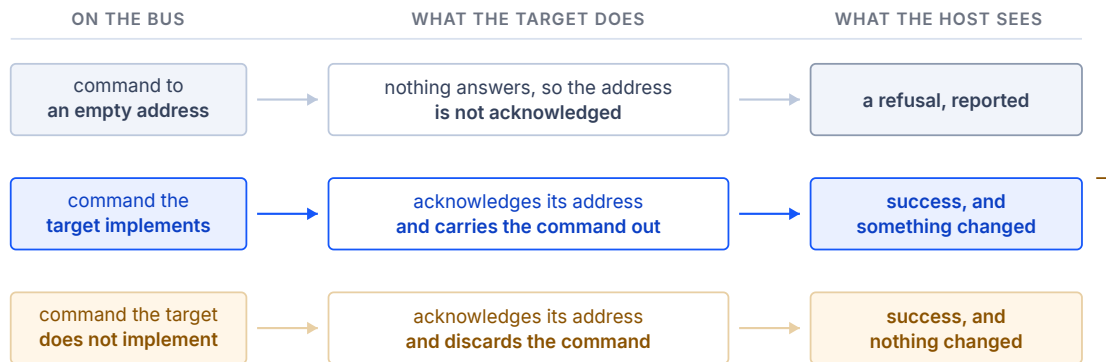
That is not the adapter concealing an error. The same commands sent to an address with no target on it are refused, and the refusal is reported, as Table 8 shows.

Table 8. The same commands at the target and at an unoccupied address

Command	At the target, <code>0x08</code>	At an unoccupied address, <code>0x0B</code>
GETPID	<code>SUCCESS</code> , returns the identifier	<code>I3C_NACK_ADDRESS</code>
GETBCR	<code>SUCCESS</code> , returns <code>0x06</code>	<code>I3C_NACK_ADDRESS</code>
SETMWL	<code>SUCCESS</code>	<code>I3C_NACK_ADDRESS</code>
ENTAS1	<code>SUCCESS</code>	<code>I3C_NACK_ADDRESS</code>
Private read	<code>SUCCESS</code> , returns data	<code>I3C_NACK_ADDRESS</code>

The absence of a refusal is therefore a real observation about the target, not a gap in the controller's reporting. The target acknowledged its address and discarded the command.

Three outcomes of one command, and what the host can tell apart



The result code separates row A from the other two. It does not separate B from C.

Only an observable effect does: a value that changes when set, an address that stops answering, a counted interrupt rate. Row A is the control that proves refusals are reported at all, which is what makes the absence of one in row C meaningful.

Figure 1. Three outcomes of one command, and which of them the result code can tell apart.

It follows that support cannot be established from a result code, and has to be established from an observable effect. Three forms of evidence are used in this document:

- **set and read back.** Send SETMWL with a value, then GETMWL. If the value does not change, the part does not implement the pair. This gives a definite negative.
- **an effect with a negative control.** Send SETNEWDA to move the target's dynamic address, then confirm the target answers at the new address and no longer answers at the old one. The second half is what makes it evidence.

- **a counted rate.** Enable an interrupt and count arrivals over a known window, then compare against the rate the sensor was configured to produce.

Where no observable effect can be constructed, the honest verdict is that support is undetermined. That is not the same as unsupported, and this document does not report it as such. The activity state commands are the clearest case: entering an activity state is intended to change the target's power consumption, and no current measurement was available on this bench, so those rows read undetermined and say why.

NOTE a value returned by a command the part does not implement carries no information. GETMWL returns `[0, 0]` on the BMI323, `[80, 80]` on the BMP581 and `[50, 50]` on the BMP585, and none of the three implements it. The evidence is that the value does not change when set, not what the value is.

5.6 Support matrix

Table 9 is produced by the `features` command of the supplied utility, which runs the behavioral tests described above and reports one of three verdicts per row. It is reproduced here from the runs recorded in Section 8.

One row needs its test stated, because the obvious test for it gives the wrong answer. A group address is a write destination: it exists so a controller can address several targets at once, which means a read from one has no defined answer and a target may refuse it while implementing group addressing correctly. Asking whether the target answers a read at the group address therefore cannot distinguish a part that lacks the feature from one that has it. The test used here writes a register through the group address and reads that register back at the target's own dynamic address, and requires three results: the write must be refused before SETGRPA, land while the group address is assigned, and be refused again after RSTGRPA. The trial value is first written at the target's own address to confirm the register will hold it, so that a refused group write cannot be confused with a register that ignores the value.

All three parts refuse the write at every stage, at three different group addresses, while SETGRPA itself returns a successful result each time. This is the same pattern Section 5.5 describes and not an additional finding.

Table 9. Measured common command code support

Command	BMI323	BMP581	BMP585	Evidence
ENTDAA	Supported	Supported	Supported	Target enumerated with the expected identifier
RSTDAA	Supported	Supported	Supported	Table cleared, old address then refuses
SETNEWDA	Supported	Supported	Supported	Address moved, old address refuses
GETPID, GETBCR, GETDCR	Supported	Supported	Supported	Returned the datasheet values
ENEC, DISEC	Supported	Supported	Supported	Interrupts start and stop, repeatably
SETXTIME, GETXTIME	Supported	Supported	Supported	A reported byte changed after the command
SETMWL, SETMRL	Not implemented	Not implemented	Not implemented	Value unchanged after being set
GETMWL, GETMRL	Not implemented	Not implemented	Not implemented	As above
SETGRPA, RSTGRPA	Not implemented	Not implemented	Not implemented	A write through the group address is refused before, during and after assignment
HDR modes	Not supported	Not supported	Not supported	BCR bit 5 clear
GETHDRCAP	Not implemented	Answers 0x00	Answers 0x00	No HDR modes reported
Hot-join	Not observed	Not observed	Not observed	No hot-join request at any point
GETSTATUS	Undetermined	Undetermined	Undetermined	Answers, not independently checked
GETMXDS	Undetermined	Undetermined	Undetermined	Answers, no observable effect
ENTAS0 to ENTAS3	Undetermined	Undetermined	Undetermined	Needs a current measurement
RSTACT	Undetermined	Undetermined	Undetermined	No observable effect from the host

SETXTIME warrants a note, because it is the one row in Table 9 where repeating the test does not repeat the result. Its sub-commands select modes and the mode bits latch, so a sub-command the part is already in is correctly a no-op. On the BMI323, from a GETXTIME of `03 00 0D 78`, sub-command `0x3F` moves the second byte to `0x01` and never moves it again, while `0xDF` moves it to `0x03`. A host that sends one sub-command and sees no change cannot conclude the command is unimplemented.

Enabling the timing control feature also changes the interrupt payload. On the BMI323 the in-band interrupt payload grows from one byte to four and bit 7 of the mandatory byte, which the specification reserves for its own use, is set. The three added bytes were nearly constant across a run and are not a per-interrupt counter; what they encode was not established. The mode persists across a host restart and a bus reset, and is cleared by a soft reset.

6 Procedure

6.1 Enumerate and identify

```
python i3c_mems.py identify --device bmp585
```

```

bmp585 (Bosch Sensortec, barometric pressure sensor)
  dynamic address 0x08

  PID 07 70 10 51 10 00
    MIPI member id (47:33)      0x03B8
    id type selector (32)       0
    device id (31:16)           0x1051
    instance id (15:12)         0b0001
    reserved (11:0)             0x000

  BCR 0x06
    device role                  I3C target
    advanced capabilities (bit 5) no, so SDR only
    virtual target support       no
    offline capable              yes
    IBI mandatory payload        yes
    IBI capable                  yes
    max data speed limit         no limit

  DCR 0x62  pressure sensor, per the MIPI DCR registry

  CHIP_ID register 0x01 reads 0x51, raw 51
    chip id field 0x51, expected 0x51: match

  register access: 1-byte data, 0 dummy byte(s) before a read payload

```

6.2 Configure a measurement

The BMI323 is configured with a single write to ACC_CONF at `0x20`. The value `0x3127` selects an accelerometer range of ± 8 g, an output data rate of 50 Hz, and an enabled power mode.

The BMP581 and BMP585 require an order rather than a single write. Configuration registers are written while the part is in standby, and only then is the measurement mode entered:

1. write ODR_CONFIG at `0x37` with a power mode of standby
2. write OSR_CONFIG at `0x36` with `press_en` set, enabling pressure alongside temperature
3. write ODR_CONFIG again with the desired output data rate and a power mode of normal
4. read OSR_EFF at `0x38` and confirm `odr_is_valid`, which reports whether the rate and oversampling combination was accepted

IMPORTANT on the BMP581 and BMP585 the interrupt block adopts a new INT_CONFIG value only at the next transition from standby into a measurement mode. The register accepts the write at any time and reads back the new value immediately, so a host that configures interrupts while the part is measuring sees every register correct and no interrupts. The effect measured on a BMP585 configured for 10 Hz was: interrupt configuration written in normal mode, no interrupts in 2 s; the same registers followed by a standby-to-normal transition, 21 interrupts; configured in standby and then entering normal, 20 interrupts. The datasheets' own guidance covers this: write configurations before switching into the measurement mode.

6.3 Read data

```
python i3c_mems.py stream --device bmp585 --seconds 2 --interval 0.5
```

Pressure is the 24-bit unsigned value divided by 64, in pascals. Temperature is the 24-bit signed value divided by 65536, in degrees Celsius. The accelerometer axes are 16-bit signed values divided by 4096 counts per g at the ± 8 g range.

The reading is self-checking, which matters because a plausible-looking number from a misconfigured part is the hardest error to notice. At rest the BMI323 reports an acceleration magnitude of 1.006 g against a configured range whose scale factor is 4096 counts per g, so the range field and the decode agree. Ambient pressure read 1003.52 hPa on the BMP585 and 1003.88 hPa on the BMP581 in the same room within the hour, agreeing to 0.36 hPa across two parts on two boards.

6.4 Raise the sensor interrupt on the bus

The BMI323 routes an interrupt source to one of three destinations. Field `acc_drdy_int` in INT_MAP2 at `0x3B` occupies bits 11 and 10 and takes `0b01` for the INT1 pin, `0b10` for INT2 and `0b11` for the I3C in-band interrupt. Writing `0x0C00` selects the last of these, and the interrupt then arrives over the two wires already present, with no interrupt pin connected.

The order that works was established step by step, each step measured over a one-second window, and is given in Table 10.

Table 10. What enables an in-band interrupt on the BMI323

State	Interrupts in 1 s
Sensor disabled	0
DISEC sent, so the target's interrupt is disabled	0
Sensor running, no routing configured	0
INT_MAP2 written, routing configured, still disabled	0
Controller configured to accept interrupts, still disabled	0
ENEC sent with the ENINT bit	50

ENEC is what enables the interrupt. Neither the routing write nor the controller's own configuration does so.

```
python i3c_mems.py ibi --device bmp585 --seconds 3 --show 3
```

```

bmp585 at 0x08: interrupt routed to the I3C IBI, pulsed
  IBI from 0x08 MDB 01 data ready=True, FIFO full=False, FIFO threshold=False, pressure
out of range=False
  IBI from 0x08 MDB 01 data ready=True, FIFO full=False, FIFO threshold=False, pressure
out of range=False
  IBI from 0x08 MDB 01 data ready=True, FIFO full=False, FIFO threshold=False, pressure
out of range=False

31 interrupts in 3.02 s = 10.28/s
configured output data rate 10 Hz (+2.8%)

```

Each interrupt carries a mandatory byte identifying its cause, and the bit assignment is per part. Table 11 gives the two layouts.

Table 11. Mandatory byte assignment, data-ready shown in bold

Bit	BMI323	BMP581 and BMP585
0	FIFO watermark or FIFO full	Data ready
1	Sample ready, accelerometer, gyroscope and temperature	FIFO full
2	Feature engine interrupts	FIFO threshold
3	Reserved	Pressure out of range
5:4	Interrupt group identifier	Zero
7	Defined by the specification	Zero

A data-ready interrupt therefore arrives as `0x02` from the BMI323 and `0x01` from the two pressure sensors. There is no shared layout to rely on.

IMPORTANT the BMP581 and BMP585 latch their interrupt by default. INT_CONFIG at `0x14` has a reset value of `0x35`, in which `int_mode` selects latched, and INT_STATUS at `0x27` is clear-on-read. The interrupt therefore asserts once and is not raised again until the host reads INT_STATUS. A host that does neither receives exactly one interrupt and then silence. Both remedies were measured at a configured 10 Hz: selecting pulsed mode gives 9.96 interrupts per second, and leaving the part latched while reading INT_STATUS after each interrupt gives 10.00. The BMI323 does not behave this way, so a host written against it and then pointed at a BMP58x will appear to find a broken interrupt.

`int_en` in the same register enables the physical interrupt pin and does not gate the in-band interrupt. With the bit clear the measured rate was 10.00 interrupts per second, and with it set, 9.95.

6.5 Disable the interrupt and reset

DISEC with the DISINT bit disables the target's interrupt. The BMI323 datasheet asks for DISEC to be sent before RSTDAA, so that the target cannot raise an interrupt during the address reset, and the supplied utility follows that order.

NOTE a single DISEC does not always stop the stream. Over repeated runs of the feature battery on a BMI323 at 50 Hz, roughly one attempt in six left the stream still arriving, while 45 consecutive attempts at 25, 50 and 100 Hz with no other traffic on the bus never failed. The cause was not established. A host should confirm that the stream has stopped rather than assume the command took effect; a second attempt was always sufficient.

Some target state survives both a host restart and a bus reset. The interrupt enable set by ENEC is one example, and the timing control mode described in Section 5.6 is another. A soft reset returns the part to its documented reset state: `0xDEAF` written to CMD at `0x7E` on the BMI323, `0xB6` on the BMP581 and BMP585.

```
python i3c_mems.py reset --device bmi323
```

7

Reference

Table 12. Utility commands

Command	Purpose
<code>scan</code>	Enumerate the bus and report what answered
<code>identify</code>	Decode a target's identity registers
<code>read</code> , <code>write</code>	Read or write a register, with a read-back after a write
<code>stream</code>	Print decoded samples by polling
<code>ibi</code>	Route the sensor interrupt onto the bus and count what arrives
<code>features</code>	Establish what the target implements, from observable effects
<code>rates</code>	Find the highest error-free bus rate
<code>reset</code>	Soft reset the part and re-enumerate

Table 13. Session options, accepted before or after the command

Option	Default
<code>--serial</code>	First adapter found
<code>--voltage</code>	3300
<code>--push-pull</code>	<code>PUSH_PULL_5_MHZ_50_DC</code>
<code>--open-drain</code>	<code>OPEN_DRAIN_1_MHZ</code>
<code>--drive</code>	<code>FAST_MODE</code>
<code>--device</code>	Required for all commands except <code>scan</code>
<code>--latched</code>	Off, so pulsed. BMP581 and BMP585 only

Table 14. Registers used in this document

Part	Register	Address	Use
BMI323	CHIP_ID	0x00	Identity, low byte
BMI323	ACC_DATA_X to Z	0x03 to 0x05	Acceleration
BMI323	ACC_CONF	0x20	Range, rate, power mode
BMI323	INT_MAP2	0x3B	Interrupt routing, including to the IBI
BMI323	CMD	0x7E	Soft reset, 0xDEAF
BMP58x	CHIP_ID, REV_ID	0x01, 0x02	Identity
BMP58x	CHIP_STATUS	0x11	Interface mode and I3C error flags
BMP58x	INT_CONFIG	0x14	Latched or pulsed, pin enable
BMP58x	INT_SOURCE	0x15	Interrupt source enable
BMP58x	TEMP, PRESS data	0x1D to 0x22	Measurements
BMP58x	INT_STATUS	0x27	Interrupt status, clear-on-read
BMP58x	OSR_CONFIG, ODR_CONFIG	0x36, 0x37	Oversampling, rate, power mode
BMP58x	OSR_EFF	0x38	odr_is_valid
BMP58x	CMD	0x7E	Soft reset, 0xB6

8

Measured results

8.1

Test configuration

Table 15. Test configuration

Item	Value
Adapter	Binho Supernova, hardware revision B, firmware 4.4.0
SDK	binhosupernova, Python 3.12, Windows 11
Accessory	Binho I3C Target Board, BIN103
Bus	3.3 V, push-pull 5 MHz at 50% duty cycle, open drain 1 MHz, fast-mode drive strength
Enumeration	RSTDAA followed by ENTDA
Targets	BMI323, BMP581, BMP585 on Bosch Shuttle 3.0 boards, one at a time

8.2 Results

Support. The matrix in Table 9 was produced by the `features` command against each part. The verdict counts were 11 supported, 7 not implemented and 7 undetermined for all three parts, and each part was measured twice to confirm the counts are reproducible.

Interrupt rate against configured rate. The interrupt rate tracks the sensor's configured output data rate.

Table 16. Interrupt rate against configured output data rate

Part	Configured	Measured	Window
BMI323	50 Hz	50.00 per second	3.0 s
BMP581	10 Hz	10.00 per second	4.0 s
BMP585	10 Hz	9.96 per second	4.0 s
BMP585	30 Hz	29.75 per second	4.0 s

NOTE the rate is comparable with the configured rate. Individual arrival times are not. Measured over 149 consecutive intervals at 50 Hz, the gaps between arrivals ranged from 0.00 to 47.00 ms with a mean of 20.03 ms, and 6 of the 149 were under 1 ms, because the host transport delivers interrupts in batches. No jitter or latency figure is given in this document for that reason.

Bus rate. The method is that a rate passes only if a fixed loop of 120 reads of the chip identifier register completes with no error and no incorrect value. The two rates are not independent: the adapter rejects some combinations as an invalid frequency pair, so the open-drain sweep was run at the highest push-pull rate that passed rather than at the default.

Both parts measured passed every rate the adapter offers, to a push-pull rate of 12.5 MHz at 50% duty cycle and an open-drain rate of 4.17 MHz.

IMPORTANT this is a register read loop, not a throughput measurement. It establishes that the parts operate without error at the top of the adapter's range under the stated conditions. It does not characterise sustained transfer performance, and no such conclusion should be drawn from it.

9

Troubleshooting

Table 17. Troubleshooting

Symptom	Probable cause and action
Nothing enumerates	Shuttle board not seated, or the target board's power jumper not selecting a rail. Confirm the rail before assuming a bus fault
Chip identifier appears wrong on a BMI323	The register is 16 bits and its upper byte is reserved, reading <code>0x1143</code> . Compare the low byte against <code>0x43</code>
The identifier differs between two reads after enumeration	The first reads after ENTDA are not settled. Discard two reads, as the datasheets prescribe. Only PID bit 12 is affected
Exactly one interrupt arrives, then none	A BMP581 or BMP585 in its default latched mode. Select pulsed mode, or read INT_STATUS after each interrupt
Interrupt registers are correct and no interrupt arrives	On a BMP58x, the interrupt block adopts INT_CONFIG only at the next standby-to-normal transition. Configure in standby, then enter the measurement mode
Interrupts arrive before they were enabled	The target's interrupt enable survives a host restart. Send DISEC, or soft reset, at the start of a session
Interrupts stop unexpectedly on a BMP58x	After an S0/S1 error the part may stop transmitting until the next start or stop condition. A dummy read to any device provides one
A command returns success and nothing changes	Expected. See Section 5.5. Establish support from an observable effect
A single DISEC does not stop the stream	Confirm the stream has stopped and repeat. See Section 6.5
The adapter rejects a rate pair	The open-drain and push-pull rates are not independent. Raise the push-pull rate or lower the open-drain rate
A configuration write is lost	On a BMP58x, several registers accept writes only in standby. Write configuration before entering a measurement mode

10

Acronyms**Table 18.** Acronyms

Term	Meaning
BCR	Bus characteristics register
CCC	Common command code
DAA	Dynamic address assignment
DCR	Device characteristics register
ENTDAA	Enter dynamic address assignment
HDR-DDR	High data rate, double data rate
IBI	In-band interrupt
IMU	Inertial measurement unit
MDB	Mandatory data byte
MEMS	Micro electro-mechanical system
ODR	Output data rate
OSR	Oversampling ratio
PID	Provisional identifier
RSTDAA	Reset dynamic address assignment
SDR	Single data rate

11

Note about the source code in the document

Example code shown in this document and supplied in the accompanying archive is provided for evaluation and reference. It is released under the MIT license, the text of which is included in the archive.

Table 19. Contents of the assets archive

File	Description
<code>AN0004.md</code>	Markdown source of this document
<code>assets/i3c_mems.py</code>	Reference utility, shared with AN0005
<code>assets/LICENSE</code>	MIT license covering the example code

12

Revision history

Table 20. Revision history

Revision	Date	Description
1.0	31 August 2026	Initial release.
1.1	31 August 2026	Section 5.6: the SETGRPA and RSTGRPA row is re-measured with a write through the group address rather than a read from it, and the test is now stated. The verdict is unchanged for all three parts. Section 8.2: the BMI323 verdict counts are corrected to 11 supported, 7 not implemented and 7 undetermined. The previous counts were taken from a run in which the part was still holding a mode left behind by an earlier run, which reported SETXTIME as undetermined; the utility now clears that state and the counts are reproducible across runs.

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